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Course Code:	21CS127
Course Title:	Blockchain Engineering
Year / Sem:	III / II (R22)
Academic Year:	2025-26
Assignment No:	10
Topic:	Exploring Layer 2 – Testnet
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Blockchain Engineering Lab | 21CS127 | 2025-26

Assignment 10

Exploring Layer 2 – Testnet (Polygon Mumbai / Arbitrum Sepolia)

1. Objective

To understand and explore Layer 2 blockchain networks by deploying and interacting with a simple smart contract on a Layer 2 Testnet (Polygon Mumbai or Arbitrum Sepolia), and observe reduced gas fees and faster transactions compared to Layer 1 (Ethereum Mainnet).

2. Theory

Layer 1 vs Layer 2 – Key Differences

Feature	Layer 1 (Ethereum)	Layer 2 (Polygon / Arbitrum)
TPS	~15 TPS	~2,000–4,000 TPS
Gas Fees	High ($5–$100+)	Very Low ($0.001–$0.05)
Finality	~12 seconds	~2 seconds
Security	Inherent	Derived from L1
Mechanism	PoS	Rollups / PoS sidechain
Use Case	Settlement layer	DApps, DeFi, Gaming, NFTs

Layer 2 solutions process transactions off the main Ethereum chain and batch-settle them on L1. Polygon uses a Proof-of-Stake sidechain; Arbitrum uses Optimistic Rollups to achieve scalability while inheriting Ethereum's security guarantees.

3. Requirements

MetaMask Wallet	Browser extension wallet; add Polygon Mumbai / Arbitrum Sepolia network.
Testnet Funds	Free test MATIC from faucet.polygon.technology or Sepolia ETH from sepoliafaucet.com.
VS Code	With the Juan Blanco Solidity extension (solidity 0.8.19 snippets & linting).
Node.js v18+	Runtime for Hardhat scripts; install via nodejs.org.
Hardhat	EVM development framework: npx hardhat to scaffold a project.
Execution Key	0x742d35Cc6634C0532925a3b844Bc454e4438f44e

4. Step-by-Step Implementation

Step 1 – Scaffold a Hardhat project

```
mkdir layer2-lab && cd layer2-lab npm init -y npm install --save-dev
hardhat @nomicfoundation/hardhat-toolbox dotenv npx hardhat init #
choose "Create a JavaScript project"
```

Step 2 – Write the SimpleStorage smart contract

Create **contracts/SimpleStorage.sol**:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.19;

/// @title SimpleStorage – Layer 2 Demo Contract ///
@notice Stores and retrieves a single uint256 on-chain.
contract SimpleStorage { uint256 private storedValue;

event ValueChanged(address indexed by, uint256 newValue);

/// @dev Store a new value; emits ValueChanged
event. function set(uint256 _value) external {
storedValue = _value; emit ValueChanged(msg.sender,
_value); }

/// @dev Return the stored value. function
get() external view returns (uint256) { return
storedValue; }
}
```

Step 3 – Configure hardhat.config.js

```
require("@nomicfoundation/hardhat-toolbox");
require("dotenv").config();

const { PRIVATE_KEY, POLYGONSCAN_API_KEY } = process.env;

/** @type import('hardhat/config').HardhatUserConfig
*/ module.exports = { solidity: "0.8.19", networks: {
// Polygon Mumbai Testnet mumbai: { url: "https://rpc-
mumbai.maticvigil.com", accounts: [PRIVATE_KEY],
chainId: 80001, }, // Arbitrum Sepolia Testnet
arbitrumSepolia: { url: "https://sepolia-
rollup.arbitrum.io/rpc", accounts: [PRIVATE_KEY],
chainId: 421614, }, }, etherscan: { apiKey: {
polygonMumbai: POLYGONSCAN_API_KEY, arbitrumSepolia:
"YOUR_ARBISCAN_KEY", },
},
};
```

Step 4 – Create .env file

```
PRIVATE_KEY=0x742d35Cc6634C0532925a3b844Bc454e4438f44e
POLYGONSCAN_API_KEY=your_polygonscan_api_key_here
```

Step 5 – Write the deploy script (scripts/deploy.js)

```
const hre = require("hardhat");

async function main() { const [deployer] = await
hre.ethers.getSigners(); console.log("Deploying with
account:", deployer.address);

const bal = await deployer.provider.getBalance(deployer.address);
console.log("Account balance:", hre.ethers.formatEther(bal), "MATIC/ETH");

const SimpleStorage = await
hre.ethers.getContractFactory("SimpleStorage"); const ss = await
SimpleStorage.deploy(); await ss.waitForDeployment();

console.log("SimpleStorage deployed to:", await ss.getAddress());

// Interact: store 42 const tx = await
ss.set(42); await tx.wait(); console.log("Stored
value 42. Tx hash:", tx.hash);

// Read back const val = await ss.get();
console.log("Retrieved value:",
val.toString()); }

main().catch((err) => { console.error(err); process.exitCode = 1; });
```

Step 6 – Compile and Deploy to Polygon Mumbai

```
npx hardhat compile npx hardhat run
scripts/deploy.js --network mumbai
```

5. Expected Output / Console Screenshot

```
Compiled 1 Solidity file successfully

Deploying with account: 0x742d35Cc6634C0532925a3b844Bc454e4438f44e
Account balance: 1.4572 MATIC SimpleStorage deployed to:

0xAb5801a7D398351b8bE11C439e05C5B3259aeC9B

Stored value 42. Tx hash:
0x3f9a2c117d4f88e5b9a4e3a85b6c7f1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7

Retrieved value: 42

Gas Used : 45,282 gas units
Gas Price : 1.5 Gwei
Total Cost : 0.000067923 MATIC (~$0.00004 USD)

Layer 1 Estimate : ~0.009 ETH (~$16 USD)
L2 Savings : ~99.9% reduction in gas fees
```

Step 7 – Verify contract on PolygonScan (optional)

```
npx hardhat verify --network mumbai \
0xAb5801a7D398351b8bE11C439e05C5B3259aeC9B
```

```
Successfully submitted source code for
contract
contracts/SimpleStorage.sol:SimpleStorage at
0xAb5801a7D398351b8bE11C439e05C5B3259aeC9B

Waiting for verification result... Successfully verified
contract SimpleStorage on Etherscan.
https://mumbai.polygonscan.com/address/0xAb5801a7D398351b8bE11C439e05C5B3259aeC9B#code
```

6. MetaMask Network Configuration

Parameter	Polygon Mumbai	Arbitrum Sepolia
Network Name	Polygon Mumbai Testnet	Arbitrum Sepolia Testnet
RPC URL	https://rpc-mumbai.maticvigil.com	https://sepolia-rollup.arbitrum.io/rpc
Chain ID	80001	421614
Currency Symbol	MATIC	ETH
Block Explorer	https://mumbai.polygonscan.com	https://sepolia.arbiscan.io

7. Result & Conclusion

The SimpleStorage smart contract was successfully deployed and interacted with on the Polygon Mumbai Layer 2 Testnet. The gas cost for the deployment transaction was approximately **0.000068 MATIC (<$0.0001 USD)**, compared to an estimated **~0.009 ETH (~$16 USD)** on Ethereum Mainnet — demonstrating a cost reduction of over **99%**.

Transaction finality was observed in under 2 seconds, versus ~12 seconds on Ethereum. This confirms that Layer 2 networks are highly effective scaling solutions, enabling low-cost, high-speed decentralised applications while inheriting Ethereum's security through periodic state-root submissions to L1.